

Binary Search.

let
arr $\rightarrow$

start		mid		end
2	4	11	20	37
0	1	2	3	4

 $\rightarrow$ sorted array.

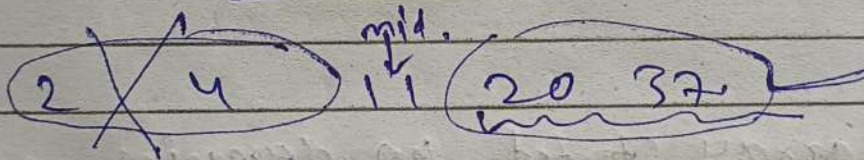
target = 20.

B.S bolta hai starting or end ko add karke or 2 se divide kar do.

$$B.S = \frac{start + end}{2} = mid.$$

Ab mid Jo hai wo target ke equal nhi hai.

$$[20 \neq 11]$$



to 20, 11 se bada hai toh wo 11 ke right side of exits karega. matlab ~~right~~ ^{right} half me exits karega or ~~left~~ left half of discard kar denge.

i.e.

```
if (arr[mid] == target) {  
    return mid;  
} else if (arr[mid] < target) {  
    start = mid + 1;  
} else {  
    end = mid - 1;  
}
```

}

4. Properties

- Binary search honcha sorted Array pe lagega.
- Ye bhut fast process hota hai.
- Aur Ye Ranges pe bhi work karta hai.

ex:-

	start		mid.		End
	↓		↓		↓
target = 18.	2	5	11 13	18	20
	0	1	2 3	4	5

$$\text{mid} = \frac{\text{start} + \text{end}}{2}$$

or

$$\text{mid} = \frac{\text{start} + \text{End} - \text{start}}{2} \quad \left. \vphantom{\frac{\text{start} + \text{End} - \text{start}}{2}} \right\} \text{to avoid overflow.}$$

Note:-

Agar array sorted ho decreasing order me.

ex:-

	↓		↓		↓		↓
→	8	7	54	36	22	18	1
	0		1	2	3	4	5

Tab bhi same kam karte, mid nikalte.

but

```

if (arr[mid] < target) {
    end = mid - 1;
}

```

```

else {

```

```

    start = mid + 1;
}

```

?